

Automatic Generation of a 3D Braid Hair Model from a Single Image

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Figure 1: Input images and generated hair models (©Just dance, Alter-ego, Roman Samborskiy/Shutterstock, Inc.)

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1 INTRODUCTION

The demand for 3D human models has been increasing recently; however, creating them is difficult. In particular, hair comes in a variety of hairstyles and complex shapes, which makes it difficult to create. However, it would be easy to create various hair models if they could be automatically reconstructed from portrait images.

We aim to generate an entire hair model including braids (regular three-strand braids) from a single portrait image (Fig. 1). Although the generation of hair models from images has been achieved to some extent in previous studies [Zhang and Zheng, 2019; Saito et al., 2018], braids and dreadlocks were not successfully generated for the entire head. Our method uses an estimation network to estimate three key points, thickness, and frequency (number of braid units) of braids, and generates braids from these points based on an algorithm. Therefore, compared with existing braid restoration methods [Sun et al., 2021], our method can estimate braids even if they are not fully visible in the image. Additionally, because we generate braids based on an algorithm, they are less likely to be broken than those generated using deep learning.

To summarize, our contributions are as follows: (1) Combining an estimation model and an algorithm makes it possible to generate a hair model of the entire head with braids, which is difficult with existing methods. (2) Our method can estimate a hair model from a single image in which the braids are not fully visible.

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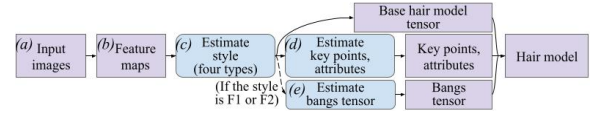


Figure 2: Workflow of our method

2 METHOD

In our method, estimation models determine the key points, thickness, and frequency of braids, and hair flow, and an algorithm generates a hair model based on them. Note that to simplify the estimation and increase accuracy, portrait images for inputs for datasets and test workflow are limited to those in which the tips of the braids are roughly visible. The outline is as follows:

Build An Estimation Model. (1) Create estimation models that estimate braid styles (four types), key points $P = P_s, P_e, P_t$ (3D coordinates of the start point, end point, and hair tip of the braid), attributes (thickness and frequency), and bang flow from a hair mask; feature maps; and a depth map created from a portrait image.

Test Workflow Below. (2) Generate feature maps from a portrait image and hair mask (Fig. 2(b)). (3) Estimate the braid style, key points, attributes, and bang flow from the feature and depth maps (Fig. 2(c, d, e)). (4) After determining the rough hair flow from the key points and a tensor of the base hair model, generate the braids and bangs parts, respectively.

2.1 Datasets

Our method uses datasets D^{st} , D^{br} , and D^{at} that estimate the style, key points, and attributes, and a dataset D^{ba} that estimates the tensor that represents the bang flow (Fig. 3). Only the ground truth is different for D^{st} , D^{br} , and D^{at} .

D^{st} , D^{br} , and D^{at} are approximately 100 sets of two types of feature maps, styles, coordinates of key points, and attributes (Fig. 3(b) top). The confidence map and orientation map [Zhang and Zheng, 2019], which are feature maps, are generated from a portrait image and hair mask (Fig. 3(a) top). The former represents the likelihood of each pixel being hair in grayscale, and the latter represents the direction of hair flow in color. The coordinates of key points are

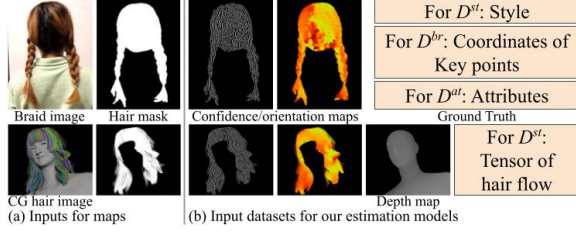


Figure 3: Datasets (top: for style, key points, and attributes, bottom: for bangs)

represented by integer grid coordinates corresponding to the braids in the image based on a $128 \times 128 \times 96$ 3D grid surrounding a head model where hair is generated. The ground truths for D^{st} , D^{br} , and D^{at} are the style type, coordinates of key points, and attributes created manually, respectively.

Unlike the other datasets, D^{ba} uses CG images instead of portrait photographs to generate maps, can accept hair images that are not braids as input, and requires a depth map of the head model. These data are created by applying approximately 350 polyline hair models [Hu et al., 2015] to the identical head model. The ground truth is a tensor of hair flow, which is the direction of hair per grid, calculated from the polyline data. To create approximately 3,000 sets, the same hair model is rendered multiple times with random head orientation, hair color, and reflectance.

2.2 Our Models

We proposed using four models to estimate the braid style, key points, attributes, and bangs [Zhang and Zheng, 2019]. We used ResNet18 [He et al., 2016] for all estimations, except for that of bangs. We normalized all inputs and outputs from 0 to 1.

We trained the model to estimate the key points on a dataset of approximately 25 sets for each style. We trained the model to estimate braid styles and attributes on a dataset of approximately 100 sets of braids for all styles.

2.3 Test Workflow

2.3.1 Inputs and Feature Map Generation. Our method uses a portrait image and hair mask as input (Fig. 2(a)). In case bangs are also generated, we also require a depth map of the head model. We initially, generate two feature maps from the input portrait and hair mask (Fig. 2(b)).

2.3.2 Estimation of the Style, Key Points, and Tensor of Bangs. Our method classifies braids into four styles based on whether the tips of the braids are in front of or behind the shoulder and whether there are one or two braids (referred to as F1, F2, B1, and B2 hereafter). First, we estimate the style from the generated feature maps (Fig. 2(c)). The estimated style determines a corresponding base hair model tensor. Next, we estimate the coordinates of the key points and the attributes of the braid (Fig. 2(d)). Note that we estimate only $P_s.x$, $P_s.y$, $P_e.y$, and $P_t.y$ because P_s is the backmost point of the head model satisfying $P_s.x$ and $P_s.y$, and P_e and P_t are consequently determined when we grow hair along the base hair model and reach the height $P_e.y$ and $P_t.y$ (2.3.3).

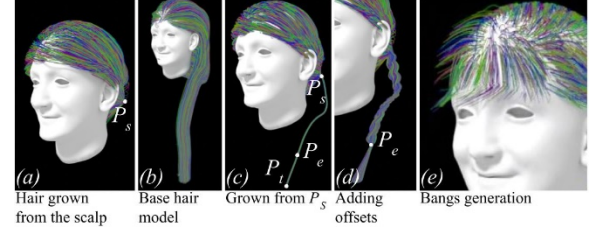


Figure 4: Hair model generation

If the style is F1 or F2, we determine that bangs are included in the image and estimate the bangs. We estimate the bangs tensor using the two feature maps and the depth map as input (Fig. 2(e)). In the case of B1 or B2, the bangs are not visible in the image and are difficult to estimate; thus, we assume an all-back style.

2.3.3 Braid and Bangs Generation. We generate a hair model based on the estimated key points and attributes. First, we gather all the hair with a total of 10,000 polylines to a single point P_s by growing them from random positions on the scalp in the shortest distance along the head model (Fig. 4(a)). For F2 and B2, there are two sets of P ; therefore, we gather them on each side of the hair. Next, we grow the collected hair from P_s to a height $P_t.y$ along the tensor of the base hair model (Fig. 4(b, c)). Then we apply an offset to these straight hairs using basically trigonometric functions and shape the braid (Fig. 4(d)). We use the attributes as coefficients. The user specifies the parameters to adjust the disorder and spread of the hair tips of the braid at this time.

If the style is F1 or F2, then we synthesize the bangs by regrowing hair from the scalp of a front part of the hair model on the basis of the tensor of the bangs estimated (Fig. 4(e)).

3 CONCLUSION AND FUTURE WORK

In this study, we generated an entire hair model with braids. Tied hairstyles are difficult to generate with existing learning-based methods, however, our method can be used for ponytails, in addition to other braids, knotted curly hair, and frizzy hair by changing the function that provides the offset.

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